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WEEK – 7,8

[*] daa.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX 100

struct Node {
    char ch;
    int freq;
    struct Node *left, *right;
};

struct Node* createNode(char ch, int freq) {
    struct Node* node = (struct Node*)malloc(sizeof(struct Node));
    node->ch = ch;
    node->freq = freq;
    node->left = node->right = NULL;
    return node;
}

void swap(struct Node** a, struct Node** b) {
    struct Node* temp = *a;
    *a = *b;
    *b = temp;
}

void sort(struct Node* arr[], int n) {
    int i;
    for(i=0;i<n-1;i++) {
        int j;
        for(j=i+1;j<n;j++) {
            if(arr[i]->freq > arr[j]->freq)
                swap(&arr[i], &arr[j]);
        }
    }
}

void printCodes(struct Node* root, int code[], int top) {
    if(root->left) {
        code[top] = 0;
        printCodes(root->left, code, top+1);
    }

    if(root->right) {
        code[top] = 1;
        printCodes(root->right, code, top+1);
    }

    if(!root->left && !root->right) {
        printf("%c : ", root->ch);
        int i;
        for(i=0;i<top;i++)
```

[*] daa.c

```c
        printf("%c : ", root->ch);
        int i;
        for(i=0;i<top;i++)
            printf("%d", code[i]);
        printf("\n");
    }
}

struct Node* buildTree(struct Node* arr[], int n) {
    while(n > 1) {
        sort(arr, n);

        struct Node* left = arr[0];
        struct Node* right = arr[1];

        struct Node* newNode = createNode('$', left->freq + right->freq);
        newNode->left = left;
        newNode->right = right;

        arr[0] = newNode;
        arr[1] = arr[n-1];

        n--;
    }
    return arr[0];
}
int main() {
    char text[MAX];
    printf("Enter a string: ");
    scanf("%s", text);

    int freq[256] = {0};
     int i;
    for(i=0;text[i];i++)
        freq[text[i]]++;

    struct Node* arr[MAX];
    int size = 0;

    for(i=0;i<256;i++) {
        if(freq[i] > 0) {
            arr[size++] = createNode(i, freq[i]);
        }
    }
    struct Node* root = buildTree(arr, size);
    int code[MAX];
    printf("\nHuffman Codes:\n");
    printCodes(root, code, 0);

    return 0;
}
```

OUTPUT:

```
Enter a string: hello

Huffman Codes:
e : 00
h : 01
o : 10
l : 11
```